Project Title	**Credit Portfolio Segmentation and Risk Pattern Analysis**
Skills take away From This Project	**Python, Pandas, Data Cleaning, EDA, Clustering, K-Means, Hierarchical Clustering, DBSCAN, PCA, t-SNE, UMAP, Data Visualization, Anomaly Detection, Streamlit, Git, GitHub**
Domain	**Finance and Banking – Loan Analytics and Credit Risk Analysis**

Problem Statement:

Financial institutions deal with borrowers who have different financial backgrounds, credit scores, employment histories, and loan requirements. Understanding these differences can help them identify borrower groups and analyse their loan portfolios more effectively.

This project aims to use **unsupervised learning techniques to group borrowers with similar financial and credit characteristics**. It will identify distinct borrower segments and analyse their key features, such as income, credit score, loan amount, and loan purpose.

The project will also **compare historical default rates across the identified borrower segments** to understand how repayment outcomes differ between groups.

The final outcome will be an interactive dashboard that presents borrower segments, their characteristics, historical default patterns, and useful business insights.

Business Use Cases:

- Borrower Segmentation
- Default Pattern Analysis
- Loan Portfolio Risk Profiling
- Anomaly Detection
- Customer Profiling
- Personalized Loan Product Design
- Geographic Portfolio Analysis
- Portfolio Trend Monitoring

Approach:

1. **Data Collection and Understanding** – Load the Lending Club dataset and understand the financial, credit, employment, and loan-related features.
2. **Data Preprocessing** – Handle missing values, duplicates, inconsistent values, and outliers; remove irrelevant identifiers and select suitable features.
3. **Exploratory Data Analysis (EDA)** – Analyse feature distributions, correlations, borrower characteristics, and default patterns.

4. **Feature Engineering and Scaling** – Encode categorical variables, transform skewed features where appropriate, and scale numerical features for clustering.
5. **Dimensionality Reduction** – Apply PCA for feature reduction and use t-SNE or UMAP to visualize borrower patterns.
6. **Clustering Analysis** – Implement K-Means, Hierarchical Clustering, and DBSCAN to identify distinct borrower segments.
7. **Cluster Evaluation and Interpretation** – Evaluate clustering quality using suitable metrics and profile each cluster based on its financial and credit characteristics.
8. **Default Pattern Analysis** – Compare historical default rates across clusters without using the `Default` column to form the clusters.
9. **Anomaly Detection** – Investigate unusual borrower profiles identified through density-based clustering and feature-level analysis.
10. **Business Insights and Visualization** – Present borrower segments, cluster comparisons, default patterns, and actionable insights through an interactive dashboard.

Results:

- Distinct Borrower Segments identified based on financial, credit, employment, and loan characteristics.
- Cluster Visualization using PCA, t-SNE, or UMAP to explore borrower patterns.
- Clustering Performance Comparison across K-Means, Hierarchical Clustering, and DBSCAN.
- Default Pattern Insights through comparison of historical default rates across borrower segments.
- Anomaly Identification to highlight unusual borrower profiles for further investigation.

- Business Insights and Recommendations to support customer segmentation and loan portfolio analysis.
- Interactive Dashboard presenting cluster profiles, financial characteristics, default patterns, and key findings.

Project Evaluation metrics:

- **Silhouette Score** – Evaluate cluster cohesion and separation.
- **PCA Explained Variance** – Assess the amount of information retained after dimensionality reduction.
- **Cluster Interpretability** – Evaluate whether the identified borrower segments have meaningful characteristics.
- **Default Rate Comparison** – Compare historical default rates across clusters for business insights.
- **Clustering Algorithm Comparison** – Compare the borrower segments produced by K-Means, Hierarchical Clustering, and DBSCAN.

Technical Tags:

Python, Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Data Preprocessing, Feature Engineering, Feature Scaling, Categorical Encoding, Unsupervised Learning, K-Means Clustering, Hierarchical Clustering, DBSCAN, Principal Component Analysis (PCA), t-SNE, UMAP, Silhouette Score, Anomaly Detection, Borrower Segmentation, Financial Data Analytics, Streamlit Dashboard

Data Set : loans_granting_model_dataset

Format: CSV (Comma-Separated Values)

Variables: Annual income, debt-to-income ratio, loan amount, credit score (FICO), employment length, loan purpose, home ownership, geographical information, loan issue date, and default status.

Data Set Explanation:

The dataset contains historical loan applicant and borrower information, including financial characteristics, credit profiles, employment details, and loan purposes. It can be used to discover borrower segments with similar characteristics and examine differences in historical default patterns.

The following preprocessing steps are required:

- Handle missing values, duplicate records, and inconsistent data.
- Remove irrelevant identifiers such as loan ID from clustering features.
- Convert date fields into suitable formats and extract useful time-based features where appropriate.
- Encode categorical variables such as loan purpose, employment length, and home ownership.
- Clean and transform numerical variables, and scale them before applying clustering algorithms.
- Exclude the default status from clustering features and use it separately to compare historical outcomes across clusters.
- Inspect feature distributions and outliers before selecting the final set of clustering variables.

.

Project Deliverables:

- Source Code: Complete Python code for data preprocessing, exploratory data analysis, clustering, and dimensionality reduction.
- Exploratory Data Analysis Report: Visualizations and insights into borrower financial profiles and loan characteristics.
- Clustering Analysis Report: Implementation and comparison of K-Means, Hierarchical Clustering, and DBSCAN.
- Dimensionality Reduction Visualizations: PCA, t-SNE, and UMAP plots to explore borrower segments.
- Cluster Profiling Report: Description of identified borrower segments and their key characteristics.
- Default Pattern Analysis: Comparison of historical default rates across the discovered clusters.
- Interactive Dashboard: Streamlit application presenting cluster profiles, visualizations, and business insights.
- Project Documentation: Problem statement, methodology, evaluation metrics, findings, limitations, and business recommendations.
- Final Presentation: Summary of the project workflow, key findings, and conclusions.

Project Guidelines:

- Data Quality: Handle missing values, duplicates, inconsistent entries, and outliers appropriately.
- Data Preprocessing: Encode categorical features and scale numerical variables before clustering.
- Feature Selection: Exclude identifiers and the `Default` column from clustering features; use `Default` only for post-clustering analysis.
- Algorithm Implementation: Implement and compare K-Means, Hierarchical Clustering, and DBSCAN using suitable parameters.

- Dimensionality Reduction: Apply PCA, t-SNE, and UMAP appropriately for feature analysis and visualization.
- Evaluation: Use suitable clustering metrics, such as the Silhouette Score, along with cluster interpretability and business relevance.
- Coding Standards: Follow PEP 8 guidelines, use meaningful variable names, modular functions, and clear comments.
- Version Control: Use Git and GitHub to maintain source code, track changes, and document project progress.
- Reproducibility: Set random seeds where applicable and document preprocessing steps, model parameters, and dependencies.
- Visualization: Create clear, properly labelled plots to communicate cluster characteristics and findings.
- Documentation: Maintain a README file describing the problem statement, dataset source, setup instructions, methodology, and results.
- Ethical Data Usage: Protect sensitive information and avoid making unsupported conclusions about individual borrowers based on cluster membership.
- Final Delivery: Submit the source code, analysis report, visualizations, project documentation, and interactive dashboard.

Timeline:

The project must be completed and submitted **within 10 days from the assigned date**

References:

Project Live Evaluation	**Project Live Evaluation**

EDA Guide	Exploratory Data Analysis (E…
Capstone Explanation Guideline	Capstone Explanation Guidel…
GitHub Reference	How to Use GitHub.pptx
Project Submission Guidelines	GitHub Project Submission G…
EDA	EDA_INDEPTH
Streamlit recording (English)	Special session for STREAM…
Pickling	PICKLING
	File

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims

to enhance comprehension and provide guidance to overcome challenges effectively.

Timing: Monday-Saturday (3:00PM to 4:00PM)

Session Link:
https://docs.google.com/document/d/1J7z_Y2whgCtFDXixysXBNXqCuhXAYD9vkiNQ-7Q7XKs/edit?usp=sharing

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.
Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:

Monday-Saturday (11:00AM to 1:00PM)
Monday-Friday (05:00PM to 06:00PM)

Booking link : https://bit.ly/project-live-booking

Created By:	Verified By:	Approved By:
Shadiya P P	Selvamani A	Nehlath Harmain